

Week 1: Foreign Exchange (FX) Markets

FINA3020 at CUHK Business School

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Zoom for this lecture only

September 2026

- Course overview and administration
- Currencies and the medium of exchange
- FX markets, order books, and spot quotations
- Cross rates and triangular arbitrage
- In-class game: triangular arbitrage race

What Is International Finance?

International finance is all of the following:

- The study of how **financial systems** are linked across countries
- How **firms and investors** are exposed to external shocks through these links
- How **financial instruments** and policy tools are used to manage risk

Several perspectives:

- ① Firms that import, export, hedge currency risk, and borrow across borders
- ② Investors that hold global portfolios and invest directly in foreign projects
- ③ Regulators that manage capital flows, exchange rates, and financial stability

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Why Study International Finance Today?

Useful knowledge for many career paths:

- Globalization of financial markets: capital mobility, foreign direct investment, and geopolitics
- Firm-level exposure: currency risk, contract design, and supply chains
- Policy trade-offs: capital controls, FX reserves, and borrowing abroad

International finance studies how firms and investors interact

- under separate monetary systems and imperfectly integrated capital markets
- without a common global regulator
- with effects on households as savers, workers, and consumers

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We begin with transactions and then move to institutions and policy:

- Weeks 1–3: FX markets, parity conditions, exchange rate expectations, and carry trades
- Weeks 4–5: FX exposure, hedging, and multinational firms
- Week 6: Midterm exam, then balance of payments and global financial cycles
- Week 7: International banking and offshore dollar markets
- Weeks 8–9: International banks and capital markets
- Weeks 10–12: Payment systems, dollar dominance, emerging-market crises, and climate finance
- Week 13: Final exam

Course Structure, Rules, and Grading

- Each class has two 75-minute halves and a 15-minute break
 - Ask questions during lecture; others likely have the same question
 - Ask lingering questions during the break while fresh in your mind
- Do **not** use phones, tablets, or laptops during lecture except for notes or in-class games
 - Please do not distract your classmates nor me. I have zero tolerance for disruptive behavior.
 - Students with approved accommodations should inform the TA
- Attendance, questions, and readings are strongly encouraged
- 10% in-class games (participation graded)
- 10% problem sets (4 assignments, submit via GradeScope)
- 5% in-class behavior
- 25% midterm exam (75 minutes in the first half of Week 6; covers Weeks 1–4)
- 50% final exam (150 minutes; cumulative)

How the Games and Problem Sets Fit Together

Most topics follow the same sequence:

- ① Learn concepts with historical and institutional context from readings and lecture
- ② Make a decision in a short in-class game before the answer is shown
- ③ Work through a longer problem set outside class
- ④ Detailed solutions released after the deadline
- ⑤ Apply the same reasoning on exams without external tools

In-Class Game: Course Survey & Icebreaker

You can discuss with neighbors, but please submit individually.

- ① Scan the QR code and choose **Course Survey & Icebreaker**
- ② Select your target career path
 - MM, IB, PE/AM, Corporate, Tech/FinTech, Government/Policy, Other
- ③ Select the international finance topic you are most eager to learn
- ④ Submit your response to test the game platform interface



[Week 1 Game Hub](#)

Participation grade based on completion

The course follows CUHK Approach 2 for AI tools:

- You may use AI tools on problem sets, but you must state which tools you used and how you used them. You are responsible for everything you submit. Check the calculations, units, definitions, and quotation conventions yourself.
- AI tools, notes, and electronic devices are not permitted during exams.
- Follow the CUHK policy on academic honesty.

I try to make the lectures narrative, engaging, and easy to follow.

- Mathematical models are still at the heart of international finance.
- A model is like a map: it simplifies the world so that we can reason about it.
- We will use algebra, calculus, and regression at the level required by the course prerequisites.

Problem sets have 3 styles of questions: multiple choice, short answer, and multi-step problems where you apply course concepts to calculations based on real-world situations

Due to the time limit on the exams, the questions are mainly multiple choice, but I strongly encourage you to learn the concepts well enough to excel at multi-step problems. Reality is complicated.

Syllabus Section 1: FX Markets and Parity Conditions

Concepts: Spot, forward, order books, triangular arbitrage, CIP, UIP, PPP, and FX regimes.

- Triangular Cross-Rate (Zero-Spread or Mid-Rate Notation):

$$S_{A/C} = S_{A/B} \times S_{B/C}$$

- Covered Interest Parity (CIP):

$$F_{t,T} = S_t \cdot \frac{1 + i_H}{1 + i_F} \implies \frac{F_{t,T} - S_t}{S_t} \approx i_H - i_F$$

- Uncovered Interest Parity (UIP):

$$\mathbb{E}_t \left[\frac{\Delta S_{t+T}}{S_t} \right] \approx i_H - i_F$$

- Real Exchange Rate (RER):

$$Q_{F/H,t} = \frac{S_{F/H,t} \cdot P_{F,t}}{P_{H,t}}$$

** Convention: $S_{F/H}$ is units of home currency per unit of foreign currency (the market pair F/H). Interest rates are matched holding-period returns.*

Syllabus Section 2: Firms and Currency Risk

Concepts: Transaction, economic, and translation exposure; forwards, money-market hedges, options, and capital budgeting.

- Unhedged versus Forward-Hedged Cash Flow:

$$CF_{H,T}^{\text{unhedged}} = CF_F S_T, \quad CF_{H,T}^{\text{forward}} = CF_F F_{0,T}$$

- Put Hedge on a Foreign-Currency Receivable:

$$CF_{H,T}^{\text{put}} = CF_F \max(S_T, K) - CF_F p_0 (1 + i_H)^T$$

- Multinational Capital Budgeting (Home-Currency NPV):

$$\text{NPV}_H = -CF_0^* S_0 + \sum_{t=1}^T \frac{\mathbb{E}_0[CF_t^* S_t]}{(1 + k_H)^t}$$

** Here p_0 is the home-currency option premium per foreign unit; k_H must price the risk in the converted cash flow.*

Syllabus Section 3: Global Flows and Intermediaries

Concepts: Balance of payments accounting, twin deficits, offshore Eurodollar markets, cross-currency basis, and capital markets.

- Balance of Payments Accounting Identity:

$$CA + KA - FA + NEO = 0$$

- Saving, Investment, and the Twin Deficits:

$$CA = S_{\text{national}} - I = (S_{\text{private}} - I) + (T - G)$$

- One-Period Cross-Currency Basis (CIP Wedge):

$$b_{H/F} = \frac{F}{S}(1 + i_F) - 1 - i_H, \quad b_{H/F} = 0 \text{ under CIP}$$

- Unhedged Foreign Return in Home Currency:

$$R_{\text{home}} = (1 + R_{\text{foreign}})(1 + g_S) - 1 \approx R_{\text{foreign}} + g_S, \quad g_S \equiv \frac{S_T - S_0}{S_0}$$

** IMF BPM6 convention: FA is net acquisition of financial assets minus net incurrence of liabilities; NEO is net errors and omissions. Basis signs vary with the quoted leg.*

Syllabus Section 4: Regulation and Current Topics

Concepts: Payment rails and settlement, US dollar dominance, sudden stops, sovereign debt dynamics, blended finance, and carbon markets.

- Cross-Border Payment Cost (Economic Decomposition):

$$\begin{aligned}\text{Economic Cost} \approx & \text{Explicit Fees} + \text{FX Spread} \\ & + \text{Prefunding Liquidity Cost} + \text{Expected Settlement-Loss Cost}\end{aligned}$$

- Sovereign Debt Dynamics:

$$d_t = \frac{1 + r_t}{1 + g_t} d_{t-1} - \text{pb}_t + \text{SFA}_t$$

- Blended-Finance First-Loss Rule:

$$L_{\text{first loss}} = \min(L, F), \quad L_{\text{senior}} = \max(0, L - F)$$

where F is public first-loss capital and L is the project loss.

** Note: This is an introductory course preview. You do not need to memorize these equations today; we will derive them step by step during the semester.*

- Money is a generally accepted medium of exchange, greatly reducing the need for a double coincidence of wants. It also serves as a unit of account and store of value.
- A currency is a monetary denomination or system (such as HKD). Notes and coins are only part of money; bank deposits also function as money.



A generally accepted medium of exchange replaces direct barter

Source: Steemit



Hong Kong banknotes are issued by designated commercial banks under HKMA arrangements

Source: HKMA

In many societies before the 19th century, commodity money included precious-metal coins, while small-denomination coins often used base metals.

- A coin's commodity value depended on its weight and purity; face value also reflected legal-tender rules and seigniorage.
- Merchants could compare foreign coins by metal content, subject to assay costs and official valuation rules.



Silver dollars from Spain and China

Source: Barrons

A Table of COINS, as they now pass in the following Places.

	England.			Philadelphia N. York.			
	l.	s.	d.	l.	s.	d.	
English Sixpence,	0	0	6	0	0	9	0 0 9
English Crown,	0	5	0	0	7	6	0 8 0
French Crown,	0	5	0	0	7	6	0 8 0
Guinea,	1	1	0	1	14	0	1 16 0
Spanish Pistole,	0	16	0	1	7	0	1 9 0
French Pistole,	0	16	0	1	6	6	1 8 0
Moydore,	1	7	0	2	3	6	2 6 0
Johannes,	3	12	0	5	15	0	6 6 0
Half Johannes,	1	16	0	2	17	6	3 3 0
Doubloon,	3	6	0	5	8	0	5 16 0
Spanish Dollar,				0	7	6	0 8 0
Pifteen,				0	1	4	0 1 6

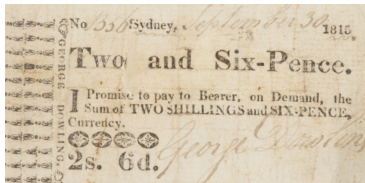
N.B. All the coins of Spain Silver are sold in London, by the ounce, and often varies, but seldom or ever exceeds 48. 5d.

Silver dollar conversion table in colonial America

Source: Old Stone Fort Museum

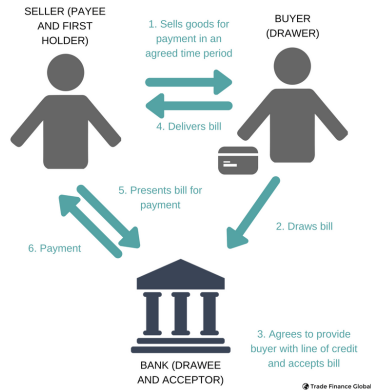
Circulating claims and promissory instruments helped shape modern banknotes.

- Many early banknotes were redeemable promises to pay coins or bullion, although not every promissory note circulated as money.
- Transferable notes reduced the cost and risk of transporting metal when recipients trusted the issuer and accepted the claim.



Promissory note from Sydney, Australia (1815)

Source: Dixon Library



Later in the course, we study international trade finance

Source: Trade Finance Global

China had the earliest documented paper-money systems, but early versions were not proper currencies

- Tang Dynasty (around 800 AD): “Flying cash” transferred funds over distance. It was mainly used for remittances vs daily use.
- Song Dynasty: The government took over issuance in Sichuan in 1023. Later over-issuance and weakened redemption contributed to depreciation and inflation.



Flying cash transferred funds without moving metal currency

Source: Qulishi

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Modern sovereign currencies are generally fiat money (fiat = *let it be done* $\approx$ *by decree*; not redeemable for a fixed amount of metal).

Value depends on demand for the currency, monetary and fiscal institutions, the state's taxing capacity, and confidence that nominal issuance will remain consistent with price stability.

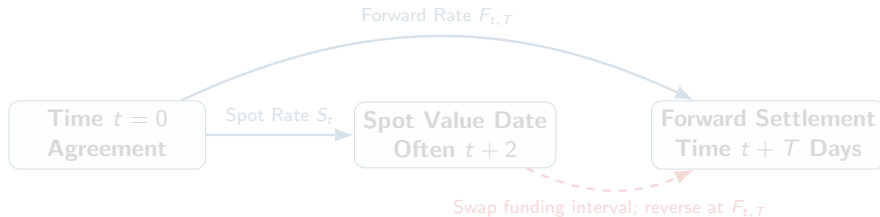
Commercial-bank deposits are also an important part of modern money creation.

All these pieces of paper are issued with as much solemnity and authority as if they were of pure gold or silver. On every piece a variety of officials write their names and put their seals. When all is prepared duly, the chief officer deputed by the Khan smears the seal with vermilion and impresses it on the paper. The money is then authentic.

(Marco Polo travelogue, observing Yuan Dynasty fiat notes under Kublai Khan)

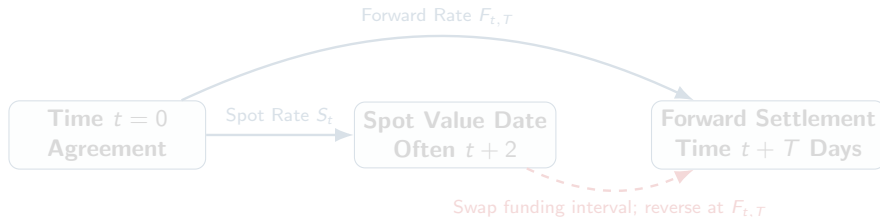
FX is the exchange of one currency for another; the trade need not cross a national border.

- Spot market = exchange for the pair's customary near-term value date (often $T + 2$, but sometimes $T + 0$ or $T + 1$)
- Forward market = contract today for a later value date at rate $F_{t,T}$; settlement may be deliverable or cash-settled
- FX swap = two opposite FX transactions agreed together for two value dates, usually a spot leg and a reversing forward leg



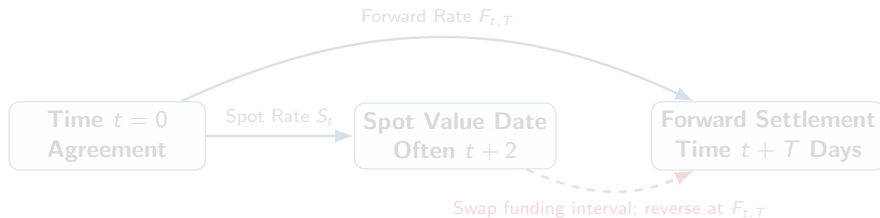
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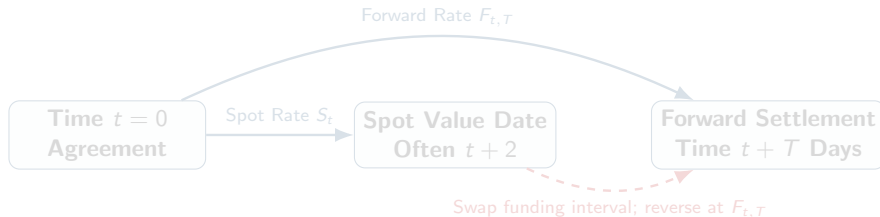
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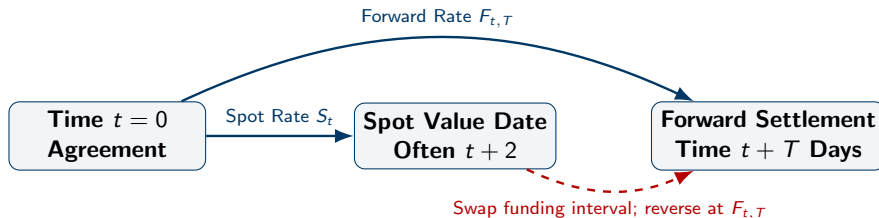
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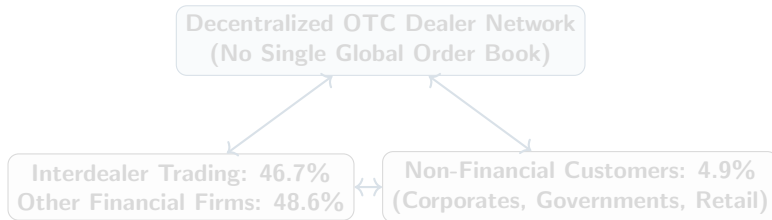


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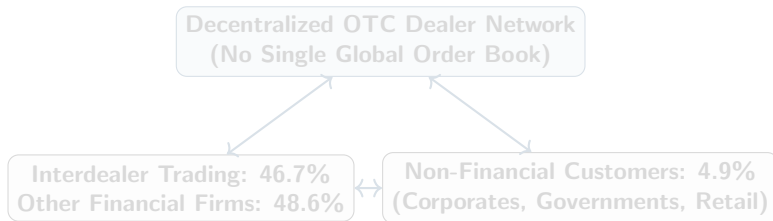


- Trading takes place 24 hours per day for 5 days per week.
- Global OTC FX turnover averaged about US\$9.5 trillion per day in April 2025.
- The US dollar was on one side of 89% of trades (currency shares sum to 200%).



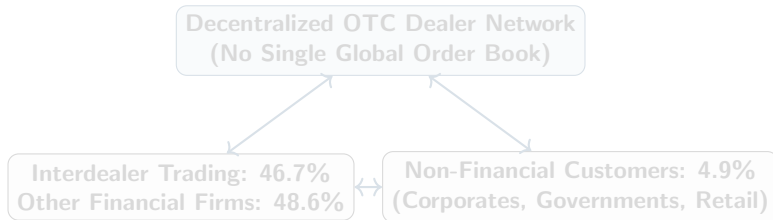
Source: BIS 2025 Triennial Central Bank Survey, final results

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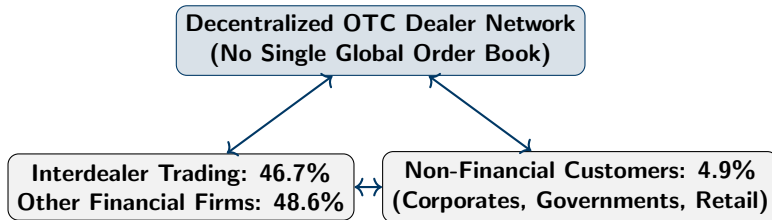
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FX market rates are written as **BASE / QUOTE = Rate**. The slash names the pair; it is not a fraction.

$$\begin{array}{ccccc} \boxed{\text{BASE Currency}} & / & \boxed{\text{QUOTE Currency}} & = & \boxed{\text{Rate}} \\ \text{Always 1 Unit} & & \text{Price in Terms} & & \\ \text{(e.g., 1 USD)} & & \text{(e.g., HKD)} & & 7.8000 \end{array}$$

- **Base Currency** (listed first): Always represents 1 unit.
- **Quote Currency** (listed second): The rate has units of Quote currency per 1 unit of Base currency.
- **Conversion Rule:** Base $\rightarrow$ Quote: Multiply by rate. Quote $\rightarrow$ Base: Divide by rate.

From Hong Kong's domestic perspective:

- **Direct Quotation (USD/HKD):** Domestic HKD price of 1 unit of foreign USD.

$$S_{\text{USD/HKD}} = 7.7800 \implies 1 \text{ USD} = 7.7800 \text{ HKD}$$

- "American terms": Direct quote from US perspective (e.g., $\text{GBP/USD} = 1.3300$).

- **Indirect Quotation (HKD/USD):** Foreign USD price of 1 unit of domestic HKD.

$$S_{\text{HKD/USD}} = 0.1285 \implies 1 \text{ HKD} = 0.1285 \text{ USD}$$

For a single mid-rate, $S_{\text{HKD/USD}} = 1/S_{\text{USD/HKD}}$.

With bid-ask quotes, invert and switch sides: $S_{\text{HKD/USD}}^{\text{bid}} = 1/S_{\text{USD/HKD}}^{\text{ask}}$ and $S_{\text{HKD/USD}}^{\text{ask}} = 1/S_{\text{USD/HKD}}^{\text{bid}}$.

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For a zero-spread mid-rate, a quote and its inverse multiply to 1. With executable dealer prices, a round-trip product above 1 creates locational arbitrage:

Dealer A bids $\text{GBP/USD} = 1.3333$ (buys GBP and pays 1.3333 USD per GBP)

Dealer B bids $\text{USD/GBP} = 0.9375$ (buys USD and pays 0.9375 GBP per USD)

- Dealer B's executable rate is much more favorable for converting USD to GBP than Dealer A's rate is for converting GBP back to USD.
- Start with 1,000 USD:
 - ① Buy GBP from Dealer B at $\text{USD/GBP} = 0.9375$: Receive $1,000 \times 0.9375 = 937.50$ GBP.
 - ② Sell GBP to Dealer A at $\text{GBP/USD} = 1.3333$: Receive $937.50 \times 1.3333 = 1,249.97$ USD.
 - ③ Riskless arbitrage profit: +249.97 USD (+25%).
- With transaction costs, no-arbitrage requires the product of executable conversion rates around any closed loop to be no greater than 1. Exact reciprocal equality applies only to a single zero-spread or mid-rate quote.

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To convert a foreign price into domestic currency, identify the Base currency:

- Example: Equipment costs 15,000,000 JPY in Japan. The direct spot quote in Hong Kong is JPY/HKD = 0.0520 (1 JPY = 0.0520 HKD).

$$\text{Domestic Cost} = 15,000,000 \text{ JPY} \times 0.0520 = 780,000 \text{ HKD}.$$

The HKD **appreciates** against foreign currency when fewer HKD are needed to buy 1 foreign unit (USD/HKD falls). Opposite is **depreciation**. Examples:

- Volcker disinflation: Tighter US monetary policy, higher relative real returns, and changing inflation expectations contributed to a broad USD appreciation.
- Asian Financial Crisis: Capital outflows and reserve losses made Thailand's peg unsustainable; after the peg was abandoned, the baht depreciated sharply.

To convert a foreign price into domestic currency, identify the Base currency:

- Example: Equipment costs 15,000,000 JPY in Japan. The direct spot quote in Hong Kong is JPY/HKD = 0.0520 (1 JPY = 0.0520 HKD).

$$\text{Domestic Cost} = 15,000,000 \text{ JPY} \times 0.0520 = 780,000 \text{ HKD}.$$

The HKD **appreciates** against foreign currency when fewer HKD are needed to buy 1 foreign unit (USD/HKD falls). Opposite is **depreciation**. Examples:

- Volcker disinflation: Tighter US monetary policy, higher relative real returns, and changing inflation expectations contributed to a broad USD appreciation.
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In-Class Game: Order Book Market-Making

Work in a team of one to three. One device per team is enough, but please list all participants' CUHK IDs

- ① Scan the QR code and choose **Order-Book Market Making**
- ② Examine the limit order book bids and asks
- ③ Submit a limit order (bid or ask) to quote market liquidity
- ④ Watch a market order fill against depth and cause slippage
- ⑤ Submit exit response: How did order size change fill price?



[Week 1 Game Hub](#)

FX is decentralized, so there is no single global book. On an electronic central limit order book (CLOB), displayed orders show amounts participants are willing to buy (bid) and sell (ask or offer) at each price.

The **tick size** is the venue's minimum price increment. A **pip** is a conventional quote unit: 0.0001 for most pairs and 0.01 for many JPY pairs. A tick may be a fraction of a pip.

The **bid-ask spread** is the difference between the lowest ask and the highest bid:

$$\text{Spread} = P_{\text{ask}} - P_{\text{bid}} > 0.$$

Liquidity providers (dealers, trading firms, etc) post limit orders. A dealer quoting both sides buys at its bid and sells at its ask.

Prices: HKD per USD Sizes: USD			
Bids (Buy USD)		Asks (Sell USD)	
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A corporate treasurer submits a market order to sell 500,000 USD for HKD.

The market order demands immediate execution against standing bids.

The order fills against the top bid (7.7990). The seller hits the bid.

Cash proceeds received:

$$\$500,000 \times 7.7990 = 3,899,500 \text{ HKD.}$$

Standing depth at 7.7990 decreases from \$1,000,000 to \$500,000.

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An importer submits a market order to buy \$100,000 USD with HKD to pay suppliers.

The market order fills against the lowest ask (7.8010).
The buyer lifts the offer.

HKD cost incurred:

$$\$100,000 \times 7.8010 = 780,100 \text{ HKD.}$$

Standing ask depth at 7.8010 decreases from \$1,200,000 to \$1,100,000.

Prices: HKD per USD Sizes: USD			
Bids		Asks	
Price	Size	Price	Size
7.7990	500,000	7.8010	1,100,000
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The market order fills against the lowest ask (7.8010).
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Standing ask depth at 7.8010 decreases from \$1,200,000 to \$1,100,000.

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7.7990	500,000	7.8010	1,100,000
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The market order fills against the lowest ask (7.8010).
The buyer lifts the offer.

HKD cost incurred:

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Standing ask depth at 7.8010 decreases from \$1,200,000 to \$1,100,000.

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An investor submits a market order to sell \$2,000,000 USD. The order size exceeds top bid depth.

The order *walks the book* across price levels:

- \$500,000 fills at 7.7990 (top bid cleared).
- \$750,000 fills at 7.7985 (second level cleared).
- \$750,000 fills at 7.7980 (\$50,000 size remains).

Average fill price:

$$\frac{500k(7.7990) + 750k(7.7985) + 50k(7.7980)}{2,000,000} = 7.79844$$

The price drop from 7.7990 is 0.0005625 HKD per USD, or about 5.6 pips when one pip is 0.0001.

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After the \$2M selling order, the best bid drops to 7.7980.

Dealers may infer that a large sell order contains information. This is an **adverse selection** problem. Dealers who bought USD also carry more long-USD **inventory risk**.

Dealers may lower both bid and ask quotes: a lower bid discourages further USD sales, while a lower ask encourages customers to buy USD from the dealers.

Dealers cancel remaining bids at 7.7980, post bids at 7.7965, and lower asks to 7.8005. The order book shifts down.

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Bids		Asks	
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A limit order instructs execution at a set price or better ($P \leq P_{\text{limit}}$ for buy orders; $P \geq P_{\text{limit}}$ for sell orders).

Many CLOBs use price–time priority, although rules vary across venues:

- **Price priority:** The highest bid and lowest ask execute first.
- **Time priority:** Orders at the same price execute in order of arrival.

Example: Dealer Y posts a bid at 7.7975 before Dealer X. An incoming sell order matches with Dealer Y first.

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- A pip equals 0.0001 for EUR/USD and GBP/USD. It equals 0.01 for USD/JPY.
- Divide the spread by the mid-price $P_{\text{mid}} = \frac{P_{\text{bid}} + P_{\text{ask}}}{2}$ to compare different currency pairs:

$$S_{\text{normalized}} = \frac{P_{\text{ask}} - P_{\text{bid}}}{P_{\text{mid}}} \times 10,000 \text{ basis points (bps)}.$$

- More liquid pairs usually have smaller spreads.
 - A major pair such as EUR/USD can trade with a spread below 1 basis point.
 - An emerging-market pair usually has a wider spread.
 - A cross such as MXN/BRL can have a still wider spread because dealers often trade through USD.
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A cross rate is an exchange rate between two currencies calculated through a third currency.

Under the Base/Quote convention, the two-leg executable cross quote through USD is:

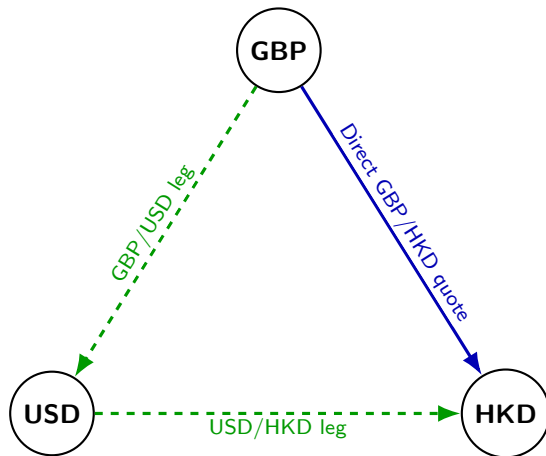
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For GBP/USD = 1.3333/1.3335
and USD/HKD = 7.8125/7.8130:

Implied GBP/HKD = 10.4164/10.4186.

A direct ask below 10.4164, or a direct bid above 10.4186, creates an arbitrage after the displayed spreads.



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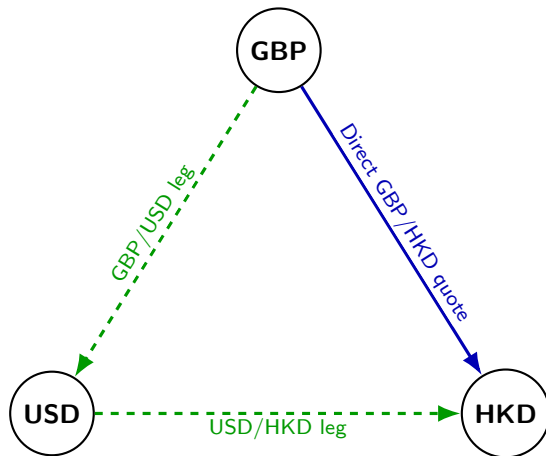
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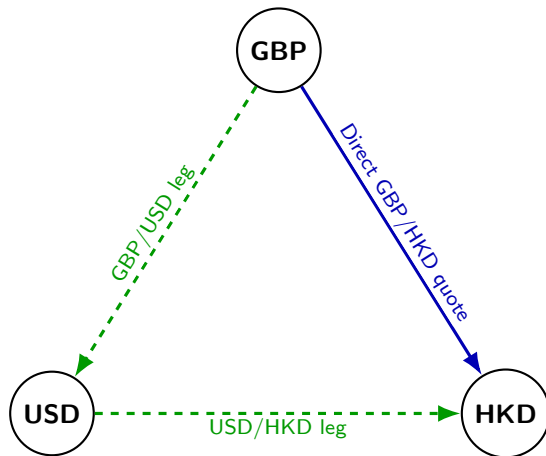
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For GBP/USD = 1.3333/1.3335
and USD/HKD = 7.8125/7.8130:

Implied GBP/HKD = 10.4164/10.4186.

A direct ask below 10.4164, or a direct bid above 10.4186, creates an arbitrage after the displayed spreads.



A cross rate is an exchange rate between two currencies calculated through a third currency.

Under the Base/Quote convention, the two-leg executable cross quote through USD is:

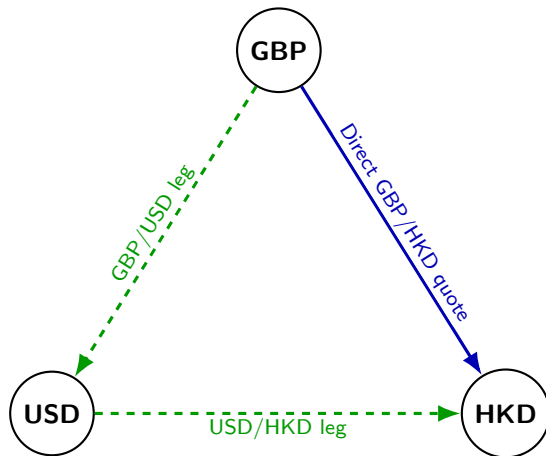
$$S_{\text{GBP/HKD}}^{\text{bid,imp}} = S_{\text{GBP/USD}}^{\text{bid}} S_{\text{USD/HKD}}^{\text{bid}}$$

$$S_{\text{GBP/HKD}}^{\text{ask,imp}} = S_{\text{GBP/USD}}^{\text{ask}} S_{\text{USD/HKD}}^{\text{ask}}$$

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Executable dealer quotes are shown as Bid/Ask. The dealer buys Base at the bid and sells Base at the ask:

- **GBP/USD:** 1.3333/1.3335
- **USD/HKD:** 7.8125/7.8130
- **GBP/HKD:** 9.999/10.000

Implied Cross-Quote through USD:

$$\text{GBP/HKD}_{\text{implied}} (\text{Bid/Ask}) = 10.4164/10.4186,$$

The direct GBP ask (10.000) is below the indirect bid (10.4164): buy GBP directly at the ask, then sell GBP for USD and USD for HKD at the two bids.

Arbitrage Result: At these made-up exchange rates, converting HKD $\rightarrow$ GBP $\rightarrow$ USD $\rightarrow$ HKD earns profit +4.164% before any costs not embedded in the quotes.

Step-by-Step 3-Leg Calculation $\rightarrow$

Available depth at the relevant side of each order book limits the arbitrage cycle:

- **GBP/HKD ask at 10.000:** 100 GBP available.
- **GBP/USD bid at 1.3333:** 75 GBP can be sold.
- **USD/HKD bid at 7.8125:** 192 USD can be sold.

Binding Constraint: The GBP/USD bid caps Leg 2 at 75 GBP. In this example, displayed depth is stated in the base currency.

Volume-Capped Profit: $75(1.3333)(7.8125) - 75(10.000) = +31.23$ HKD per cycle.

Backwards Induction Volume Proof →

Which Direction Is Profitable?

[Core Concept]

When every pair has a valid bid no greater than its ask, the two opposite cycle returns cannot both be positive.

Counter-Clockwise

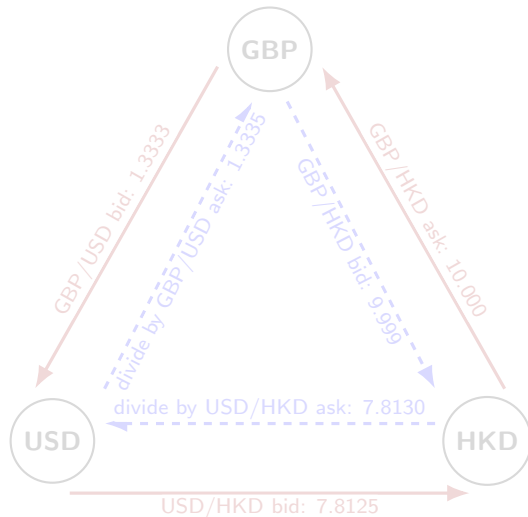
(HKD → GBP → USD → HKD): Traversing counter-clockwise buys undervalued GBP:

$$\frac{1}{10.000} \times 1.3333 \times 7.8125 - 1 = +4.16\%.$$

Clockwise (HKD → USD → GBP → HKD): The reverse cycle uses the opposite side of every quote:

$$\frac{1}{7.8130} \times \frac{1}{1.3335} \times 9.999 - 1 = -4.03\%.$$

Clockwise traversal results in a loss.



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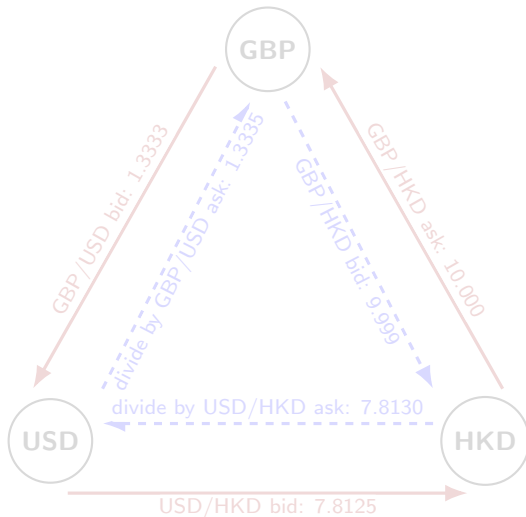
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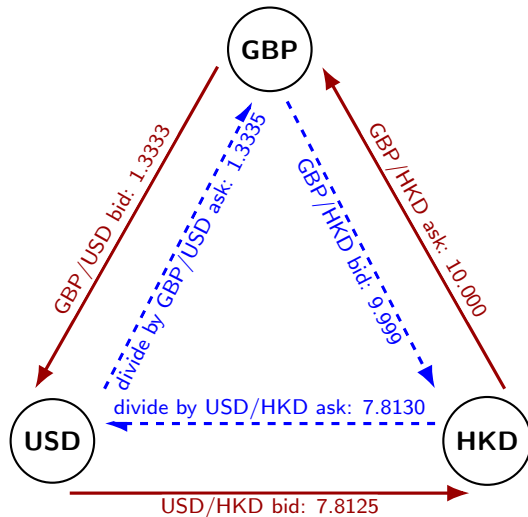
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In-Class Game: Triangular Arbitrage Race

Work in a team of one to three. One device per team is enough, but please list all participants' CUHK IDs

- 1 Choose **clockwise**, **counter-clockwise**, or **no trade**
- 2 Write down three executable rates supporting your choice
- 3 Scan the QR code, choose **Triangular Arbitrage Race**, and execute the trade
- 4 Increase the spread until the trade is no longer profitable
- 5 Submit individual exit response explaining your decision



[Week 1 Game Hub](#)

Appendix: Triangular Arbitrage Step-by-Step

Starting with 1,000 HKD:

- ① **Leg 1 (HKD → GBP):** Buy GBP at the GBP/HKD ask of 10.000.

$$\text{Receive } \frac{1,000 \text{ HKD}}{10.000} = 100.00 \text{ GBP.}$$

- ② **Leg 2 (GBP → USD):** Sell GBP at the GBP/USD bid of 1.3333.

$$\text{Receive } 100.00 \text{ GBP} \times 1.3333 = 133.33 \text{ USD.}$$

- ③ **Leg 3 (USD → HKD):** Sell USD at the USD/HKD bid of 7.8125.

$$\text{Receive } 133.33 \text{ USD} \times 7.8125 = 1,041.64 \text{ HKD.}$$

Net Profit: $1,041.64 - 1,000.00 = +41.64 \text{ HKD (+4.16\%).}$

To find the maximum executable trade size under order-book limits:

① **Leg 2 Capacity:** The binding GBP/USD bid depth allows 75 GBP to be sold.

② **Leg 1 Initial Capital Needed:** To buy 75 GBP at $\text{GBP/HKD} = 10.000$:

$$75 \times 10.000 = 750.00 \text{ HKD.}$$

③ **Legs 2 and 3:** The 75 GBP produces $75(1.3333) = 99.9975$ USD, which is then sold at the USD/HKD bid:

$$99.9975 \times 7.8125 = 781.23 \text{ HKD.}$$

Capped Profit: $781.23 - 750.00 = +31.23 \text{ HKD.}$